

AADARSH GUPTA

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EDUCATION

Indian Institute of Technology, Delhi

2019 – 2023

Bachelor of Technology in Electrical Engineering; Minor in Cognitive & Intelligent Systems

GPA: 8/10

Relevant Courses: Data Structures and Algorithms, Machine Intelligence & Learning, Computer Vision, Computational Cognition and Perception, and Control Systems.

EXPERIENCE

Founding AI Engineer

January 2026 - Present

Neatlogs

Bangalore, India

- Built production ML pipeline with **QLoRA**-finetuned **LLaMA** 3.1 (8B) for multi-label behavioral signal detection across agent traces; trained on **~16K** curated samples on A100 clusters with **MLflow**-tracked experiments, achieving **0.83–0.91 F1**; deployed on **Azure ML** and **AWS SageMaker** with blue-green rollout, load testing, and P50/P95/P99 latency benchmarking.
- Designed modular **FastAPI** services for AWS App Runner with **Kafka** consumers, powering an event-driven detection pipeline with async execution, batching, and graceful degradation across ML, regex, conditional, and PII detection (**Presidio**).
- Built AI agents for search and interpretability over complex observability traces, enabling interactive exploration and auto-generation of evaluation configs from trace context; implemented FSM-based streaming parser for structured LLM outputs and adaptive model routing with escalating reasoning budgets, achieving **sub-2s TTFT**.
- Built hybrid semantic search (**BM25 + vector**) with **RRF**-based fusion for agent trace data; benchmarked 6 embedding models across 4 vector dimensions to optimize recall–latency tradeoffs under **ClickHouse (HNSW)** retrieval constraints.
- Designed prompt template reverse-engineering pipeline using **FastCDC** chunking, **Merkle tree**-based structural diffs, and anchor-based matching; achieved **>90%** precision with **sub-50ms** extraction latency.

Software Engineer AI/ML - II

January 2025 - January 2026

Thena.ai

Bangalore, India

- Built production-grade AI agents for Thena's B2B CRM platform, integrating entity-aware memory and retrieval-augmented knowledge to enable a context-driven, agentic chat interface in under **3 months**.
- Architected AI-driven CRM workflows automating ticket deflection, dynamic summarization, and platform actions with idempotency and supersession safeguards for reliable execution during concurrent high-volume interactions.
- Engineered performance optimizations with caching and state checkpointing, achieving **85%** reduction in chat latency & **66%** faster workflow executions; enhanced **RAG** pipeline with queued background processing for high-throughput retrieval.
- Developed modular AI toolchain including **MCP server** integration, federated knowledge retrieval, web search, **NL2SQL** generation, dynamic charting, and Thena API connectors to expand agent capabilities.
- Built end-to-end evaluation and reliability infrastructure with feedback pipelines, guardrails, and observability for real-time LLM monitoring; implemented distributed testing via **GitHub Actions** and production monitoring with **Sentry**.

System Software Engineer

July 2023 - December 2024

Enphase Energy

Bangalore, India

- Developed Python-based parsers for firmware correction analysis, improving simulated power performance by **15%**.
- Deployed SciPy-based optimization scripts for control system parameter estimation through time series curve fitting.
- Implemented visualization scripts for simulation analysis; designed Python test frameworks reducing test duration by **20%**.

AI Fellow

March 2023 - May 2023

Pi School [\[Code\]](#) [\[Workshop\]](#) [\[Presentation\]](#)

Remote

- Developed an LLM-based chatbot evaluation system that emulates human assessment for feedback generation.
- Integrated assessment generation with business chatbot websites; Presented findings at a prompt engineering workshop.

Summer Intern

Medtronic: Work under patent submission [\[Report\]](#)

June 2022 - August 2022

Hyderabad, India

- Devised a novel vision mechanism for automated masking of laparoscopic surgical video; **Key Outcome:** US Patent filed
- Worked on real-time localized tracking system for *HUGO RAS* surgical robotic arms & collision-avoidance algorithms.; [\[LoR\]](#).

PUBLICATIONS AND PATENTS

[P.1] **Surgical Robotic System with Laparoscopic System.** US Patent Application — Medtronic Inc.
K V S Manoj Kumar Vadali, Sai Gautham Ravipati, **Aadarsh Gupta**

[C.1] **Text based diagnosis of COVID-19 using Data mining techniques: A comparative study.**
Aadarsh Gupta*, Aastha Valecha*, Sapna Mishra, Tapan Gandhi ** Equal contribution*
IEEE, 19th India Council International Conference (INDICON) 2022 [\[Paper\]](#) [\[Code\]](#) [\[Slides\]](#)

KEY PROJECTS

Characterizing Brain States using Deep Learning [\[Slides\]](#) [\[Thesis\]](#) August 2022 - May 2023
B.Tech Project, Advisor: Prof. Saurabh Gandhi, CSND Lab IIT Delhi

- Developed classification strategies for sleep stages from polysomnographic data using deep learning techniques.
- Synthesized neural signal processing methods to classify Lucid dream state & detection of microsleep events using EEG data.

AWARDS & ACHIEVEMENTS

AI Fellowship: Selected as 1 of 10 fellows and awarded scholarship for [Pi School of AI](#), Italy 2023

Singapore International Pre-Graduate Award: Selected as a visiting research scholar at IHPC, A*STAR 2023

Project Fellowship: Granted scholarship by [IHFC](#) for project mentored by *Addverb Technologies* 2022

JEE Advanced: Secured All India Rank 467 among 2,30,000 shortlisted candidates 2019

JEE Mains: Secured All India Rank 498 among 1.2M candidates; obtained 99.963 percentile 2019

KVPY Fellow: Awarded fellowship by Dept. of Science & Technology, Govt. of India and IISc Bangalore 2018

INChO: Awarded Certificate of Merit by IACT for securing top 1% position in NSEC 2018

NTSE: Awarded national scholarship and Certificate of Merit by NCERT, Govt. of India 2017

ROLE OF RESPONSIBILITY

Teaching Assistant January 2023 - May 2023
ELL457: Computation and Cognition IIT Delhi

- Teaching Assistant for [Prof. Sumeet Agarwal](#), for the topics in Cognition & Deep Learning, and was part of a team that assisted in creating resources and evaluation materials for over 80 students.

Teaching Assistant August 2022 - December 2022
ELL409: Machine Intelligence & Learning IIT Delhi

- Teaching Assistant for [Prof. Brejesh Lall](#), for the topics in Machine Learning & Computer Vision, and was part of a team that created quizzes, study material and question banks for over 120 students.

Mentor March 2021 - June 2021
CML101: Introduction to Chemistry IIT Delhi

- Mentored a group of 30 to help them understand concepts through regular teaching sessions and reference notes.

SKILLS

Languages & Frameworks	Python, SQL, C/C++, TypeScript, FastAPI, Next.js, React
ML/AI	PyTorch, Transformers, PEFT, QLoRA, MLflow, bitsandbytes
Databases & Systems	PostgreSQL, ClickHouse, Redis, Kafka
Cloud & Infrastructure	AWS (App Runner, SageMaker, ECS, Bedrock, S3), Azure ML, GCP (Vertex AI), Docker, Git, GitHub Actions, Grafana, Sentry